

Morgan Creek Fire

NDVI Analysis

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[3/4/26]

The Morgan Creek Fire

When?

- Started: July 9th, 2021
- Ended: October, 2021

Where?

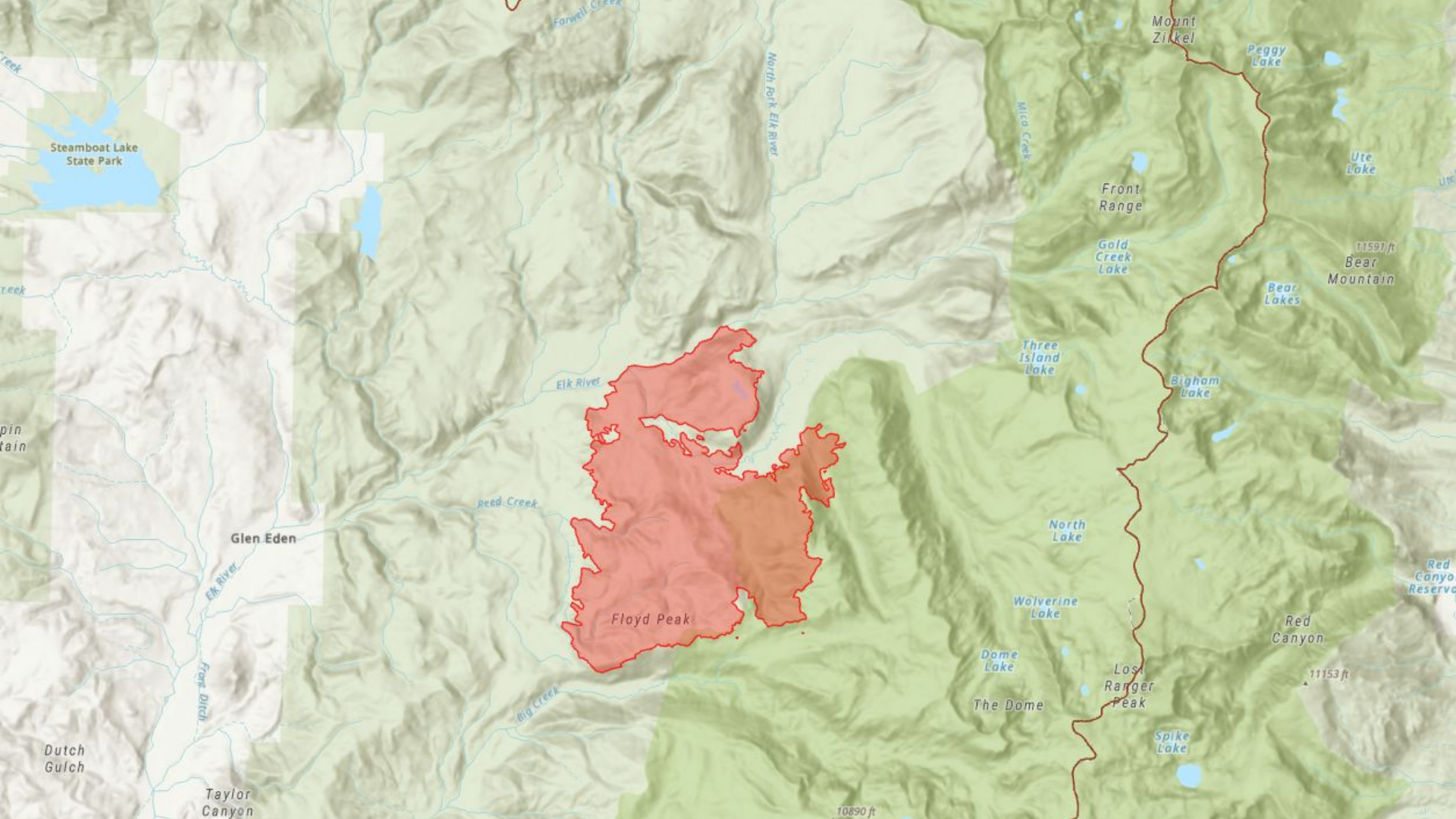
- 15 miles north of Steamboat Springs
- North Routt
 - Mount Zirkle Wilderness
 - Medicine Bow-Routt National Forest.

How?

- Lightning-strike

Grew to 4,443 acres in 10 days. and is still 0% contained with 395 firefighting personnel on scene





Study Objective

- To evaluate vegetation change before and after the Morgan Creek Fire using NDVI
- Specifically to examine whether areas near mapped beaver pond clusters show different vegetation recovery patterns compared to the surrounding burned landscape.

Analytical methods

Data Sources:

<https://earthexplorer.usgs.gov/>

Landsat 8/9

ID:
LC08_L2SP_035032_20210610_20210621_02_T1
Date Acquired: 2021/06/10
Path: 035
Row: 032



June 10th, 2021

Land Cloud Cover: 0.64

ID:
LC09_L2SP_035032_20220621_20230411_02_T1
Date Acquired: 2022/06/21
Path: 035
Row: 032



June 21st, 2022

Land Cloud Cover: 0.13

USGS [Watershed Boundary Dataset](#)

Watershed Boundary Dataset HUC 8s

Name: Watershed_Boundary_Dataset_HUC_8s
Type: Feature Layer
Owner: [esri environment](#)
Modified: 2/26/2026 11:17:14 AM
Path: <https://cmc-colorado.maps.arcgis.com/home/item.html?id=5bbefdc2511472ea9abd0afedb85c7e>



USA States Generalized Boundaries

Name: USA_States_Generalized_Boundaries
Type: Feature Layer (Hosted)
Owner: [esri dm](#)
Modified: 4/7/2025 1:25:59 PM
Path: <https://cmc-colorado.maps.arcgis.com/home/item.html?id=8c2d6d7df8fa4142b0a1211c8dd66903>



[USA Wetlands](#) - U.S. Fish and Wildlife Service

COBAM (Colorado Beaver Activity Mapper)

- Pond cluster data

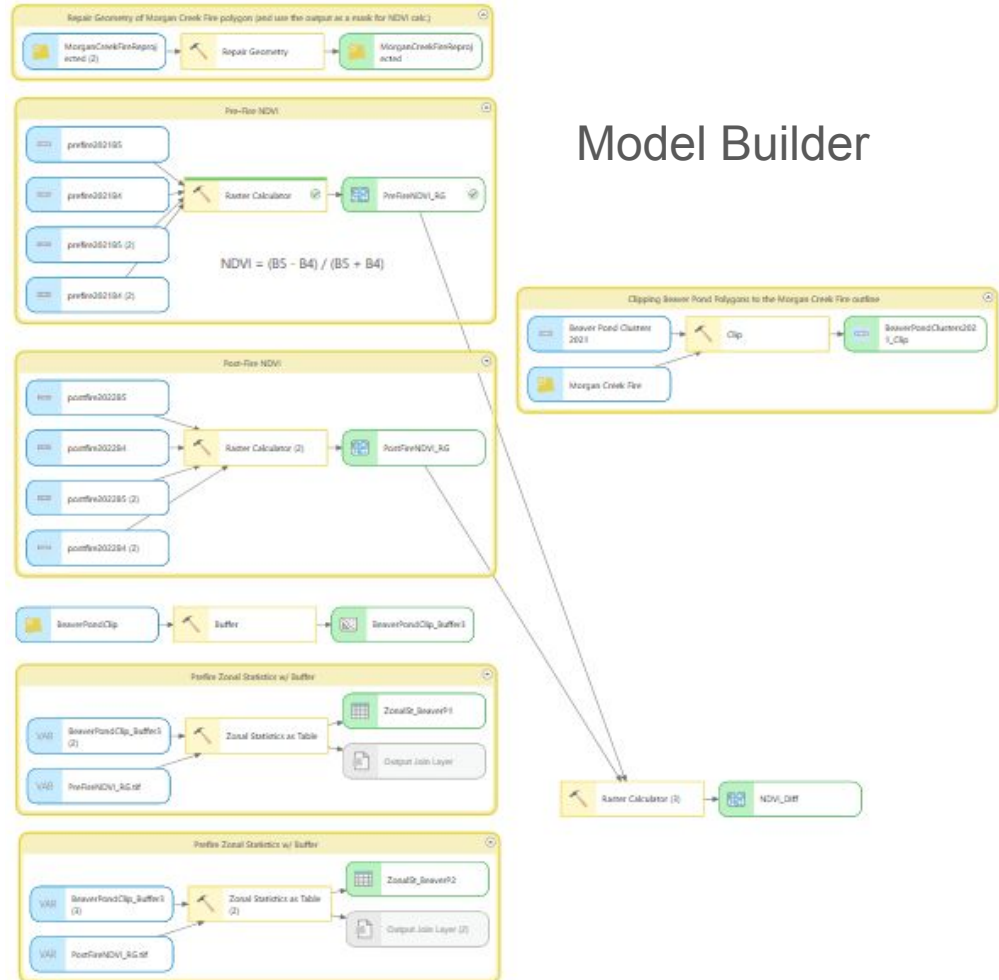
BRAT (Beaver Restoration Analysis Tool)

Methods

Geoprocessing Tools:

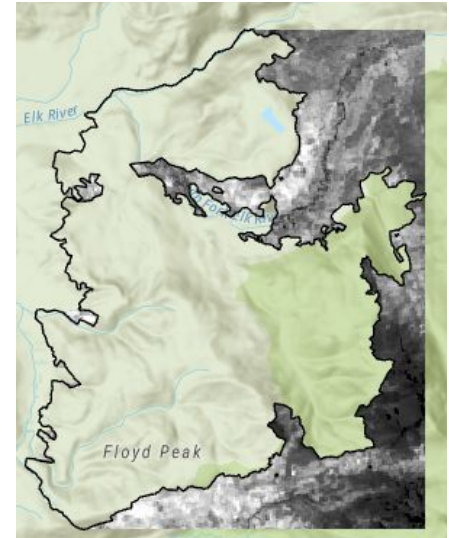
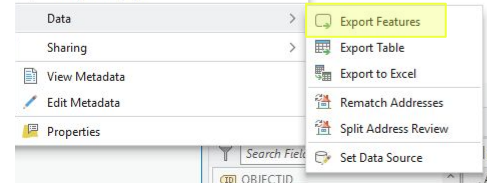
- Repair Geometry
- Raster Calculator
- Clip
- Buffer
- Zonal Statistics as Table

Model Builder



Things I Learned

- Don't create a new layer from a selected attribute (ex. To get the morgan creek fire polygon as a separate layer from the rest of the fire polygons) didn't work .
select by attribute...
 - Wrong: Selection -> make layer from selected feature
 - Instead: Data -> export feature as a layer
- Struggled with the extent and mask while calculating NDVI. Once I "Repaired Geometry" of the Morgan Creek Polygon, and re-ran "Raster Calculator" with the repaired polygon as the mask it worked!



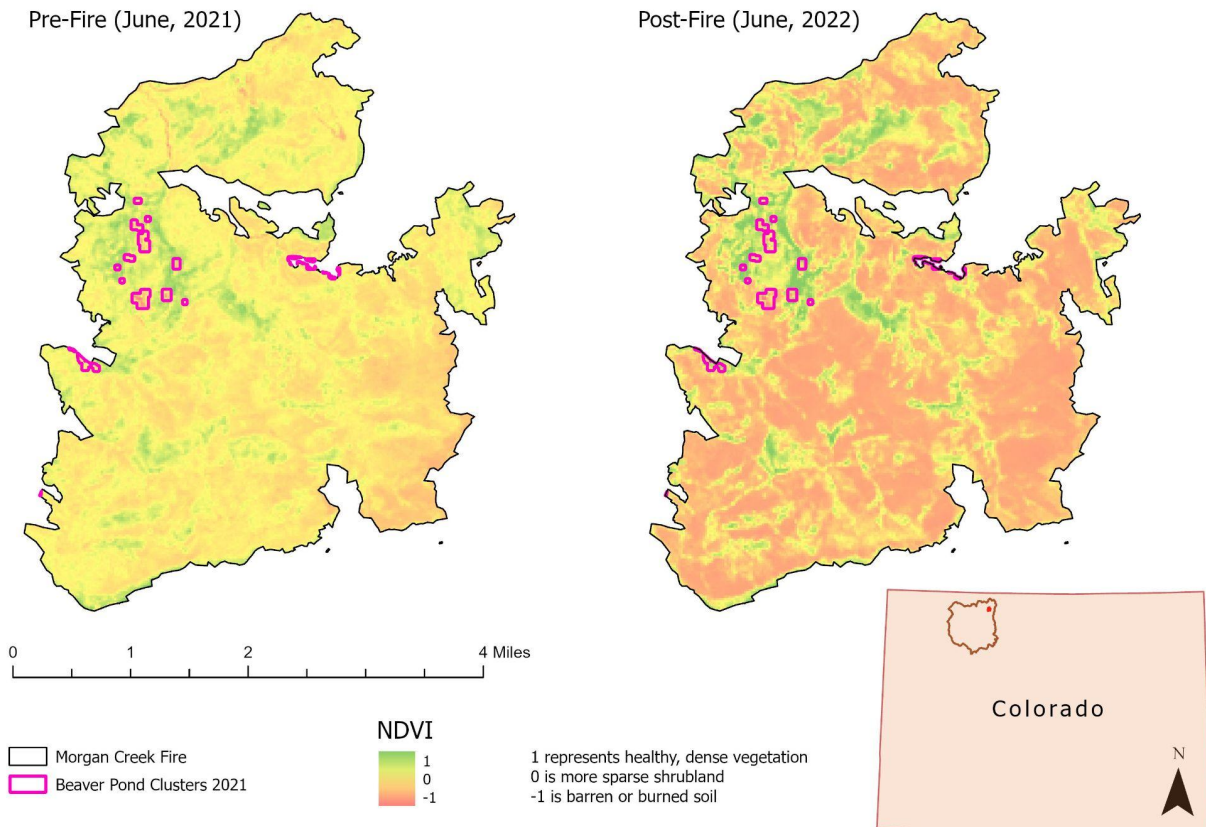
Results

Morgan Creek Fire, Routt County, CO

There is a patch of green that remains near the beaver ponds after the fire.

Within 200ft buffer zone of the pond clusters the average NDVI value changed by $+0.067$ after the fire

While the average NDVI value for the entire Morgan Creek Fire changed by -0.064 after the fire.



Discussion

Beaver ponds slow and store water, helping to maintain soil moisture.

These riparian zones appear more resilient to wildfire and may support stronger vegetation recovery.

The presence of greener vegetation patches near beaver pond clusters in the post fire NDVI imagery supports the idea that beaver influenced landscapes can contribute to ecological resilience following wildfire events.

